

IMPLEMENTATION OF AN LLM-BASED METHOD FOR EPD MATCHING IN BIM WORKFLOWS

Authors¹

¹University

The names and affiliations MUST NOT be included in the draft submitted for review.

The header consists of 10 lines with exactly 14 point spacing.

The last line below should be left blank.

Abstract

This paper presents a semi-automated workflow for assigning layer and material information based on the Industry Foundation Classes (IFC) data standard to Environmental Product Declaration (EPD) datasets from the German life cycle assessment database ÖKOBAUDAT for road infrastructure assessment. A Python-based matching tool is integrated into a Common Data Environment (CDE) via a JavaScript Object Notation (JSON) interface and addresses missing standardization through project-specific mapping of property sets. A configurable, multi-stage approach using large language models (LLM) is evaluated with respect to accuracy and cost. Pilot tests demonstrate potential for early-stage sustainability assessments.

Introduction

Germany has an extensive highway system covering approximately 13,000 km (Federal Ministry of Transport 2025). The largest agency responsible for tendering and operating highways, Autobahn GmbH, is committed to sustainability goals and e.g. aims to be 100% climate-neutral and CO₂-neutral in its construction and operation by 2045 (Autobahn GmbH). Similarly, another important public client, DEGES, is pursuing targeted digitization strategies and promoting early sustainability assessments in planning phases. (DEGES 2025)

The Federal Highway Research Institute (BAST) has published approaches for the holistic and uniform sustainability assessment of roads and other infrastructure (Oliver Fischer et al. 2016; Rainer Hess, Martina Lohmeier, Torsten Mielecke, Katharina Kunz 2022).

These are catalogs of criteria with sustainability criteria, whose assessment approaches are explained methodically and in profiles (Rainer Hess, Martina Lohmeier, Torsten Mielecke, Katharina Kunz 2022). Standardized, uniform, or holistic assessment approaches are not currently available to a comparable extent.

Furthermore, guidelines for uniform, simplified calculations of global warming potentials for road construction were developed, based on the environmental product declarations (EPD) systematics. The initial goal was to provide manufacturer-independent and transparent life cycle assessment data. (Michael Sulzbach n.d.)

The achievement of sustainability objectives in road infrastructure projects requires reliable and

comprehensive data to enable informed decision-making and life cycle-based sustainability assessments. This is particularly relevant with regard to the availability and quality of environmental data for construction materials. (Kaus and Barkmann 2025) However, EPDs required for sustainability assessment in road construction are hardly available in public databases such as ÖKOBAUDAT (Kaus and Barkmann 2025) and tend to be based on generic data. The EPDs are therefore not product-, company-, or association-specific data, and in the ÖKOBAUDAT they are commissioned by the BBSR (Federal Ministry for Housing, Urban Development and Building) and supplied by Sphera. (Federal Ministry for Housing, Urban Development and Building) Accordingly, although generic EPDs are available for sustainability assessment in early planning phases, they are to a certain extent abstract.

Ongoing research activities analyze current road construction processes and identify those planning stages that exert the greatest influence on sustainability assessment outcomes. In early design phases, geometries and material concepts are defined only at a high level, while tendering and contract award are typically based on predefined service specifications. As a result, bidders have limited opportunities to contribute to sustainability unless such criteria are explicitly addressed in the tender documents. Although functional tenders based on service programs (§7c VOB/A) shift solution development to contractors and thereby offer innovation and sustainability potential, sustainability criteria are still rarely applied as award criteria by public clients. (RWTH Aachen University 2025)

In 2024, the Council of the European Union adopted a revised Construction Products Regulation, paving the way for the future implementation of digital product passports, including their use for sustainability assessment purposes. The updated regulatory framework introduces new requirements for digital data management across the construction sector, affecting manufacturers and other economic operators alike, and aims to align regulatory practice with technological progress and long-term sustainability objectives. (Council of the EU 2024) However, while product-related environmental data are increasingly mandated at the regulatory level (Council of the EU 2024), their structured integration into planning and assessment workflows for road infrastructure remains

a significant challenge, particularly in early project phases where decisions have the greatest impact (RWTH Aachen University 2025).

Against this background, this paper aims to contribute to improving the availability and usability of environmental data for sustainability assessment in road infrastructure projects. The focus lies in supporting early planning and decision-making phases by facilitating the linkage between digital road models and environmental product information. To this end, a semi-automated, large language model (LLM)-assisted approach is proposed that enables the matching of material specifications from Industry Foundation Classes (IFC)-based road infrastructure models with EPD datasets from the German life cycle assessment database ÖKOBAUDAT. The initial objective of this approach is to evaluate technical feasibility and to establish a practical workflow suitable for direct application in future EPD developments as more EPDs become available.

State of the art

This chapter outlines and compares sustainability assessment systems as well as semi-automated and fully automated approaches for matching sustainability information. It particularly focuses on their methodological differences and on how recent studies integrate Artificial Intelligence (AI)-based methods to support or enhance automation in EPD matching.

Automation strategies for EPD matching in BIM-based LCA

Based on the reviewed literature, automation strategies for EPD matching within Building Information Modeling (BIM) -based Lifecycle Assessment (LCA) frameworks are generally categorized as either fully automated or semi-automated. Fully automated EPD matching remains uncommon in current research. Existing implementations are typically based on custom-developed plug-ins or scripts that couple with specific BIM and LCA software environments. Such workflows are most feasible when LCA functionality is embedded within BIM and relies on predefined, often generic, datasets. While these approaches offer advantages such as reduced manual effort, fewer transcription errors, and enabling rapid environmental feedback in early design stages, they are characterized by limited flexibility, strong tool dependence, and insufficient support for interoperable, model-based data exchange. The absence of standardized data structures and matching logics further constrains scalability and broader adoption. (Potrč Obrecht et al. 2020)

In contrast, semi-automated EPD matching is more frequently encountered in BIM-based LCA applications. In these workflows, material quantities are typically extracted automatically from BIM models, commonly via IFC or Bills of Materials, while the assignment and validation of LCA or EPD data remain partially manual. This approach enables the use of product-specific EPDs and adaptation to different databases, project types, and modeling conventions, though at the cost of increased time demand and reliance on expert judgment for data

selection and interpretation. Nevertheless, semi-automated workflows are often favored due to their greater methodological control and robustness when dealing with incomplete or ambiguous model information. (Potrč Obrecht et al. 2020)

Existing approaches to BIM-based EPD matching

Against this background, more recent research has investigated AI-supported approaches that extend semi-automated BIM-based LCA workflows by assisting or partially automating EPD matching. These approaches aim to reduce the remaining manual effort and uncertainty associated with semi-automated workflows.

Complementing tool-based automation efforts, several conceptual frameworks adopt a more structural perspective on EPD matching in BIM-based LCA. Studies by Wastiels and Decuypere and Horn et al. emphasize that the feasibility of EPD matching is fundamentally constrained by data availability, level of detail, interoperability, and the intended use case. Rather than prioritizing semantic automation at the tool level, approaches such as BIM2LCA focus on establishing standardized IFC-based data exchange across planning phases, linking early-stage benchmark data with detailed, product-specific LCA and EPD information in later design stages. (Wastiels and Decuypere 2019; Horn et al. 2020)

In that context, Forth et al. propose a conceptually oriented approach based on Natural Language Processing (NLP), in which IFC-based BIM elements are linked to LCA datasets using semantic similarity rather than exact string matching. This approach enables the generation of emission ranges instead of single deterministic values, thereby explicitly acknowledging uncertainty as an inherent characteristic of early-stage LCA (Forth et al. 2023).

In contrast, Hermann et al. focus on a practice-oriented implementation with the BIMetrix method, which combines semantic and geometric IFC processing with AI-supported EPD mapping within an automated ETL (Exchange, Transform, Load) workflow. While this approach demonstrates feasibility in large-scale infrastructure projects, manual verification of AI-assigned EPDs remains necessary, reflecting ongoing limitations in fully automated, reliable assignment under real-world conditions. (Hermann et al. 2024)

Petrosa et al. present an open-access, AI-driven matching tool implemented in Python that supports both semi-automatic and automatic assignment of ÖKOBAUDAT datasets. Although significant time savings are reported, the fully automated mode is explicitly described as experimental, with validation results indicating moderate matching accuracy for the fully automated approach. As a result, expert oversight remains necessary, and concerns are raised regarding transparency and computational efficiency when using general-purpose LLM. (Petrosa et al. 2025)

Challenges in AI-supported BIM-based EPD matching

The reviewed literature indicates that, despite increased interest in AI-supported BIM-based EPD matching, several challenges persist. A major structural barrier is the absence of an open-access BIM-based LCA “smart” software solution that enables automated EPD assignment within BIM workflows (Petrosa et al. 2025). Beyond software availability, the integration of EPDs into BIM workflows represents a key challenge for the LCA of infrastructure projects. In German road construction, a significant discrepancy exists between the material designations used in practice according to TL Asphalt-StB and the available EPD data sets in databases such as ÖKOBAUDAT (Kaus and Barkmann 2025). This discrepancy hinders systematic and automated EPD assignments. The insufficient standardization of IFC-based data structures, including property sets (PSETs), further complicates automated assignment (Wastiels and Decuypere 2019; Potrč Obrecht et al. 2020). Furthermore, cost and computational efficiency have been identified as key challenges of AI-assisted EPD matching. The use of LLMs imposes token-based costs that scale with the size and redundancy of the input dataset and are further constrained by token limits per request (Petrosa et al. 2025). Without adequate pre-selection mechanisms, large and heterogeneous EPD databases can therefore lead to excessive computational effort and increased Application Programming Interface (API) costs (Petrosa et al. 2025).

Methodology

Considering these challenges, the approach presented in this paper describes a methodology for semi-automatic EPD assignments. A JavaScript Object Notation (JSON)-based interface architecture allows the system to be flexibly connected to Common Data Environments (CDE). The proposed methodology specifically addresses the problem of non-standardized IFC structures and uses manual PSET assignment within the CDE to store EPD references in a layer-specific manner. To improve cost efficiency and computational robustness, the approach introduces predefined rule-based filters to preselect eligible EPDs prior to LLM-based matching. By reducing the size of the candidate dataset passed to the AI model, this strategy could lower computational effort and associated costs, while it could also improve robustness and practical applicability of AI-assisted EPD matching.

EPD matching Tool

The Python-based EPD matching tool was developed through an iterative Human-AI Collaboration (HAIC) methodology, where a domain expert (construction engineering) collaborated with an LLM (Claude Opus 4.5, Anthropic) in a conversational development process (Fragiadakis et al. 2024). This approach follows the Planning-Driven Model (PDM) identified in recent taxonomies of LLM-assisted software development which emphasizes architectural specification and human oversight (Ge et al. 2025). The implementation followed

an iterative process inspired by the PDM, as illustrated in Figure 1:

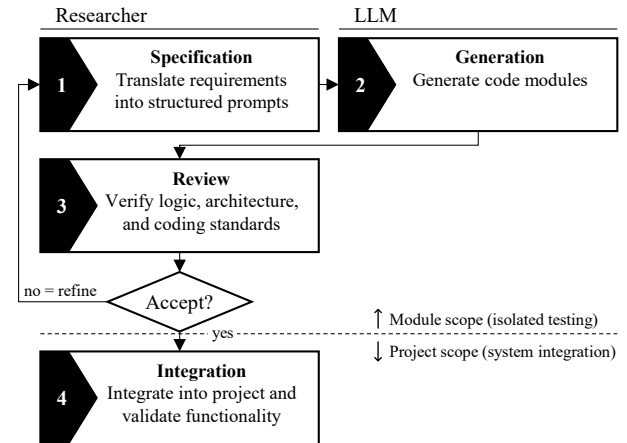


Figure 1: Iterative implementation workflow for LLM-assisted development (adapted from Ge et al. 2025)

This approach emphasizes upfront design investment to ensure structural coherence and maintainability of the resulting code (Ge et al. 2025). While “vibe coding” operates primarily at the conversational level with the AI assuming implementation responsibility (Meske et al. 2025), the proposed approach additionally enforces explicit architectural specifications and isolated module testing before project integration. Before implementing new functions, every module was tested separately to ensure it works properly before implementing it in the project.

The developed algorithm operates in five sequential processing stages, with several configurable parameters. Three of them are systematically evaluated in this study (see Table 1).

1. Context extraction

Layer-specific context information is extracted from the input data. When *EPD_PREFER_NAME_FIELD* (*NamePref*) is enabled, the layer name is given priority, as it provides semantic information about the function of the layer in the road structure. The designations in *NAME* correspond directly to EPD classifications and areas of application.

2. Material code parsing

German asphalt designations follow the nomenclature defined in TL Asphalt-StB 07/13. A typical designation such as "AC 11 D S" encodes multiple material properties.

The LLM-based approach maintains functionality even when material designations deviate from standard nomenclature or are provided in different languages. Domain-specific parsing rules are provided via *EPD_USE_GLOSSAR*, which supplies the model with TL Asphalt-StB terminology as background information. Since ÖKOBAUDAT does not yet provide sufficient information about infrastructure EPDs, the analysis of this setting excluded from the analysis.

3. EPD pre-filtering

When *EPD_USE_GLOSSAR_FILTER* (*Filter*) is enabled, a category-based filtering approach reduces the search space before LLM processing. The filtering follows a two-tier approach for asphalt materials:

- Primary matches: EPDs containing both the layer-specific term (e.g., "Binder" for binder course) and asphalt-related terms
- Secondary matches: EPDs containing only asphalt-related terms

For non-asphalt materials, the algorithm identifies the material category from predefined keywords and applies category-specific inclusion and exclusion filters. The category definitions can be extended for specific use cases.

When disabled, all EPDs from the database are passed to the LLM, increasing token consumption but potentially improving recall for edge cases.

4. Semantic matching via LLM

The filtered (or complete) EPD set is transferred to an LLM together with the material description. The prompt includes:

- Material designation and layer context,
- EPD database entries (columns: name, technical description (optional), notes (optional))
- Domain-specific matching rules
- Output format specification (JSON with confidence scores)

EPD_USE_BATCH_MODE (*Batch*) processes all layers in a single API call, significantly reducing token consumption since the EPD database and prompt are transmitted only once. Thus, it might affect accuracy.

5. Result aggregation

The LLM response is parsed and can be enriched with metadata from the EPD database. The results are sorted by confidence values.

Analysis

To isolate the contribution of individual optimizations, a full-factorial ablation study with three binary configuration switches ($2 \times 2 \times 2$) is conducted. This experimental design follows established best practices for Machine Learning ablation studies, which emphasize the importance of systematic evaluation over single-configuration comparisons (Fostiropoulos and Itti 2023; Meyes et al. 2019). The factorial design ensures that interaction effects between configuration parameters can be identified, addressing the risk of misleading conclusions from isolated component evaluations. This yields eight configurations (Table 1), which are evaluated systematically and repeated five times each. Multiple trials per configuration are essential to account for the inherent stochasticity of LLM-based matching, where identical inputs can produce varying outputs (Fostiropoulos and Itti 2023). This repetition increases the

statistical power of the analysis and enables reliable estimation of performance variance across configurations.

Table 1: Configurations

ID	NAME	Batch	Filter	NamePref
P1	Baseline	-	-	-
P2	Batch	x	-	-
P3	Filter	-	x	-
P4	NamePref	-	-	x
P5	BatchName	x	-	x
P6	FilterName	-	x	x
P7	BatchFilter	x	x	-
P8	All	x	x	x

A standardized pavement assembly with five layers is used as the benchmark case. For each layer, the most suitable EPD entry is identified via manual lookup in ÖKOBAUDAT and stored as ground truth. The selection is based on semantic correspondence between the layer specification and the EPD description, prioritizing entries that best reflect the intended layer composition. In case there is no matching EPD for a layer inside the database or semantic information stored inside the IFC models layer can be called false, then the layer is excluded from evaluation.

In addition, four different LLM models (see (Figure 2)) are considered for the semantic matching stage. Model-specific token prices are recorded as of 21 January 2026. Costs are computed from the input and output token counts per run.

EPD Matching Tool

System architecture

The developed process consists of four heterogeneous systems that communicate with each other. The EPD matching tool acts as a central communication point that merges data streams from two sources (CDE and EPD database) and forwards the enriched result to both the CDE (for model enrichment) and the LCA tool (for life cycle assessment). The process consists of the following systems:

1. **CDE:** In this project, [company] was selected as the CDE. It provides the backend processing and exposes the workflow via a user-friendly frontend dialog. [company] supplies the IFC-derived inputs (layer information, material designations, and IFC Global IDs, GUIDs). For each matching request, it generates a unique task identifier and creates a corresponding directory. The extracted data is written to a JSON file

within this directory, enabling reliable input–output association per task and concurrent usage by multiple users.

2. **EPD database:** The fundamental data source employed in this study is that provided by ÖKOBAUDAT. Subsequently, the data may be enriched with labels for the purpose of EPD use cases. Furthermore, EPDs from other sources may be appended.
3. **EPD matching tool:** The developed processing component that compares layer data from the CDE with EPD data from the database.
4. **LCA tool:** Downstream life cycle assessment tool that uses the enriched layer data, including EPD references, for environmental impact calculations.

Data exchange format

First, various options for transferring data between the systems involved were examined. For data exchange between the CDE, EPD database, and matching tool, JSON was selected over alternatives such as eXtensible Markup Language (XML) and Comma Separated Values (CSV). JSON provides compatibility with the Swagger-based REST interface used for the project's database, supports hierarchical data structures required for nested GUID arrays, and allows extensibility for additional attributes such as confidence scores. The same JSON schema is used for both the input from the CDE and the output to the LCA tool to ensure a unified data model throughout the workflow. The LCA tool receives the original layer information enriched with EPD UUIDs, enabling direct lookup of environmental impact data.

While XML offers similar structural capabilities, it introduces parsing overhead and higher resource consumption due to redundant metadata (Meleková and Kvet 2026).

CSV was excluded due to its flat structure, which cannot represent the nested relationships inherent in the data schema.

The resulting JSON is structured in the following way:

Table 2: JSON hierarchy

Attribute	Description	Example
NAME	Name of the layer group	e.g., "Deckschicht"
MATERIAL	Material specification	e.g., "AC 11 D S"
GUID	List of unique identifiers for group elements	e.g., ["202Fr\$", ...]
VOLUME	Volume associated with the group	e.g., 42.380

The JSON file groups IFC elements according to functional layer designation (NAME), material designation (MATERIAL), Global IDs (GUID), and aggregated volume. Some of the corresponding attributes are manually selected, using a dedicated user interface within the CDE.

NAME (layer designation): This study focuses on the life cycle assessment of five-layer road structures. The selected layer taxonomy (Table 3) was created in alignment with RStO (Guidelines for the Standardization of the Superstructure of Traffic Areas) (Forschungsgesellschaft für Straßen- und Verkehrswesen e. V. 2024) and refined based on IFC road models provided by the project's industry partners. That makes sure that the layer definitions reflect both normative guidance and common practice in real-world modeling workflows.

Table 3: Layer designation and associated description

Layer	NAME	Translation
1	Deckschicht	surface layer
2	Binderschicht	binder layer
3	Tragschicht	base layer
4	Schottertragschicht	unbound sub-base
5	Frostschuttschicht	protection layer

This preselection could reduce input errors and ensure consistent terminology that corresponds directly to the classifications in the EPD database. Restricting the selection to defined layer types also enables targeted pre-filtering of EPD candidates in the matching algorithm.

MATERIAL (material designation): The user visualizes the infrastructure model within the CDE and selects the layer to be evaluated by interacting directly with the 3D model. Upon selection, a properties window opens, displaying the PSETs assigned to the IFC element.

The material designation is deliberately extracted manually by the user. This is due to the previously discussed interoperability problem within the BIM workflow, that i.e. material information is stored in different PSETs depending on the authoring software, export configuration, and project convention.

Since automated extraction would be prone to errors due to this heterogeneity and could lead to incorrect EPD assignments, a semi-automated approach was chosen for this project. The user visually identifies the correct PSET within the CDE and marks the corresponding PSET in the input mask as the most relevant for the material designation.

GUID: The GUIDs are automatically extracted from the IFC model as soon as the user selects a layer. In the case of multiple selections (e.g., all elements of a surface layer over a section of a route), all GUIDs are recorded as an array. These references enable the EPD assignments to be written back to the correct model elements at a later stage.

VOLUME: The volume is automatically calculated and aggregated from the geometric properties of the selected IFC elements inside the CDE. In the case of multiple selections, the individual volumes are added together. The volume value is used in the downstream LCA tool as a reference value for extrapolating the environmental impacts based on the functional unit of the EPD.

Results

A limited set of pilot tests was conducted to validate the EPD matching tool. These tests involved an IFC models of road infrastructure with varying layer specifications (reflecting different pavement layer types and naming conventions). While the number of test cases was small due to limited access to IFC models, the preliminary results were promising. In each case, the JSON data extracted from the CDE was processed by the EPD matching tool. The outcomes were evaluated in terms of (1) matching accuracy (Figure 2) and (2) cost efficiency (Table 4).

To assess whether the multistage approach was leading to improved results, its results were compared to a basic LLM matching approach (P1) without any preprocessing.

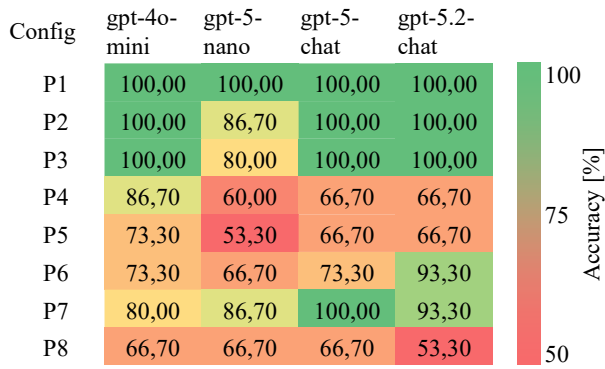


Figure 2: Heatmap of average matching accuracy per model and configuration, in %

Interestingly, the Baseline variation P1 led to 100% accuracy throughout all the used LLM models for the specific IFC model tested.

To identify the most cost-efficient configuration (Table 4), the cost per accuracy point ratio (1) was calculated as:

$$\text{Cost Efficiency} = \frac{\emptyset \text{ cost per 1000 requests}}{\text{Accuracy [\%]}} \quad (1)$$

Table 4: Cost efficiency Ranking, best to worst (Top 10)

Rank	Model Config	Accuracy [%]	Cost $\times 10^{100}$ [USD]	Cost efficiency
1	gpt-4o-mini P7	80	0,9244	0,012
2	gpt-4o-mini P8	67	0,9410	0,014
3	gpt-4o-mini P3	100	1,4120	0,014
4	gpt-4o-mini P6	73	1,3900	0,019

5	gpt-4o-mini P2	100	3,8786	0,039
6	gpt-5-nano P7	87	3,6000	0,042
7	gpt-4o-mini P5	73	3,9116	0,053
8	gpt-5-nano P2	87	4,7094	0,054
9	gpt-5-nano P8	67	4,8608	0,073
10	gpt-5-nano P5	53	5,4664	0,102

As illustrated in Table 4, the most cost-effective configuration for attaining 100% accuracy was determined to be gpt-4o-mini with configuration P3. The most inefficient configuration, which therefore did not rank within the top 10, was gpt-5.2-chat with P4, achieving an accuracy of 67% at a cost of 220,26 USD for 1,000 runs, resulting in a cost efficiency score of 3.3.

Discussion

The results indicate that cost and accuracy are strongly shaped by the choice of optimization strategy and model, rather than by model class alone.

The ablation study suggests that the proposed multi-stage workflow can reduce operational cost substantially without necessarily preserving the peak accuracy observed in the baseline. While the baseline configuration (P1) achieved 100% accuracy for the tested IFC model across all models, this should be interpreted cautiously. The benchmark comprises a single standardized pavement structure and a small number of layers. Under such conditions, the ground-truth EPD candidates may be comparatively easy to identify, and multiple EPD entries can be semantically close. Consequently, perfect accuracy in P1 may reflect the simplicity and specificity of the test case rather than general robustness. A broader benchmark set is required to confirm whether P1 remains consistently superior in accuracy when material descriptions, naming conventions, and data quality vary. In previously non systematically run tests on different pavement structures, accuracy increased, when using *EPD_USE_GLOSSAR_FILTER* (Filter). That was probably because unnecessary EPDs were excluded from the data set and therefore the LLM only had more probable options to choose from. In a small number of cases, the LLM exhibited hallucination-like behavior and returned clearly implausible assignments. For example, selecting a parquet EPD for an asphalt base layer. Therefore, the matching tool exports the five highest ranked suggestions per layer so that the user can manually select between different options inside the LCA interface. For this study though, only the highest ranked suggestions were analyzed.

The cost-efficiency ranking highlights that pre-filtering and batching change the cost-quality trade-off in different ways. The “Filter” configuration (P3) combined with gpt-4o-mini achieved 100% accuracy at the lowest cost

among the perfect-accuracy runs, resulting in a cost efficiency index of 0,014. The same model, gpt-4o-mini, achieved 100% accuracy in the baseline configuration P1 at around twelve times the cost, resulting in a cost efficiency index of 0,176. This supports the assumption that reducing the candidate set before LLM inference can lower token consumption while maintaining sufficient recall for the correct EPD. In contrast, configurations that bundle multiple layers into a single call (Batch) reduce repeated transmission of database context and therefore cut token usage per layer but may increase failure risk when the model must track multiple queries and return structured outputs for each. This effect is consistent with the observed tendency of batch-based variants to deliver competitive cost-efficiency while occasionally degrading accuracy.

Several limitations constrain the interpretability of the findings. The evaluation relies on a single pilot case IFC model and therefore a single standardized pavement assembly, and the ground truth is derived from manual selection, which introduces potential subjectivity. Especially where multiple EPDs could be considered acceptable matches. Additionally, the database coverage limitations for infrastructure materials in ÖKOBAUDAT lead to a lot of cases, where only a generic dataset can be considered and thus makes material code parsing (see Stage 2) obsolete in most cases.

Finally, model prices and behavior can change over time; cost-efficiency results should be treated as conditional on the recorded pricing date and the specific model versions used.

Overall, the study demonstrates that cost-effective EPD matching in infrastructure contexts is feasible when LLM inference is embedded into a controlled, domain-aware pipeline. The ablation results provide initial evidence that carefully designed pre-processing can preserve high accuracy while reducing cost, particularly when paired with smaller, price-efficient models.

Conclusion

This paper presented a semi-automated workflow for matching IFC-derived layer and material information to EPD datasets from ÖKOBAUDAT, implemented as a Python-based matching tool integrated into a CDE. The approach addresses a central interoperability challenge in BIM-based sustainability assessment: inconsistent IFC Property Set structures and non-standardized placement of material information across projects and authoring tools. By enabling project-specific property mapping within the CDE and applying a configurable multi-stage matching pipeline, the method reduces the time and expertise required to enrich BIM models with EPD references for downstream LCA.

The findings are based on limited pilot cases and a standardized pavement benchmark and should therefore be interpreted as an initial validation rather than a definitive performance claim. A further limitation concerns the current state of the ÖKOBAUDAT database itself: the number of available EPDs remains limited, and many matched results are generic sector averages. For the

method to reach its full potential, it is essential that more manufacturers upload product-specific EPDs. Additionally, the introduction of a well-structured, standardized labeling or classification system within ÖKOBAUDAT would greatly facilitate automated pre-filtering and improve matching precision.

Future work should expand the evaluation to a larger and more diverse set of IFC models, including additional infrastructure material categories, and investigate more rigorous ground-truth definitions (e.g., multiple acceptable EPD labels with weights or expert consensus). Moreover, improvements to prompt design and confidence calibration, as well as database-agnostic extensions to incorporate additional LCA datasets where licensing permits, represent promising directions.

In summary, the presented CDE-integrated, semi-automated matching approach demonstrates that LLM-assisted EPD matching can be used in a practical workflow. The method provides a scalable pathway for earlier and more accessible sustainability assessments by bridging digital building models and environmental databases with controllable cost and accuracy characteristics.

Acknowledgments

This research is supported by the Federal Ministry of Transport (BMV) in the funding program mFUND in the project SusInfra (Federal Ministry of Transport 2023). Thanks also to all project partners who made possible the partial results on which the study is based.

This research also benefited from BIM model data provided by DEGES, which enabled the proposed process to be refined and evaluated against practical scenarios.

AI tools (ChatGPT and DeepL) were used to support translation and to improve grammar and wording. All scientific content, interpretations, and conclusions are the authors' own.

References

- Autobahn GmbH (Ed.): Mobilität auf nachhaltigem Kurs. Available online at <https://www.autobahn.de/digitales-innovation/nachhaltigkeit/unser-weg>, checked on 1/16/2026.
- Council of the EU (Ed.) (2024): Building materials: Council adopts law for clean and smart construction products. Available online at <https://www.consilium.europa.eu/de/press/press-releases/2024/11/05/building-materials-council-adopts-law-for-clean-and-smart-construction-products/>, checked on 1/16/2026.
- DEGES (Ed.) (2025): Auszeichnung in der Kategorie „Nachhaltigkeit“: DEGES ist BIM Champion 2025. Available online at <https://www.deges.de/aktuelles/neuigkeit/auszeichnung-in-der-kategorie-nachhaltigkeit-deges-ist-bim-champion-2025/>, checked on 1/16/2026.
- Federal Ministry for Housing, Urban Development and Building (Ed.): ÖKOBAUDAT data providers.

- Available online at <https://www.oekobaudat.de/anleitungen/datenlieferanten.html>, checked on 1/16/2026.
- Federal Ministry of Transport (2023): Entwicklung eines intelligenten Nachhaltigkeitsrechners von Infrastrukturprojekten auf Basis digitaler, georeferenzierter BIM-Modelle - SusInfra. Available online at <https://www.bmv.de/SharedDocs/DE/Artikel/DG/mfund-projekte/SusInfra.html>, checked on 1/19/2026.
- Federal Ministry of Transport (Ed.) (2025): Infrastruktur. Available online at <https://bmdv.bund.de/SharedDocs/DE/Artikel/G/infrastruktur-statistik.html>, checked on 16.01.26.
- Forschungsgesellschaft für Straßen- und Verkehrswesen e. V. (2024): RStO 12/24. Richtlinien für die Standardisierung des Oberbaus von Verkehrsflächen. Available online at <https://www.fgsv-verlag.de/rsto-12-24>, updated on 2024, checked on 1/21/2026.
- Forth, Kasimir; Abualdenien, Jimmy; Borrmann, André (2023): Calculation of embodied GHG emissions in early building design stages using BIM and NLP-based semantic model healing. In *Energy and Buildings* 284, p. 112837. DOI: 10.1016/j.enbuild.2023.112837.
- Fostiropoulos, Iordanis; Itti, Laurent (2023): ABLATOR: Robust Horizontal-Scaling of Machine Learning Ablation Experiments.
- Fragiadakis, George; Diou, Christos; Kousiouris, George; Nikolaidou, Mara (2024): Evaluating Human-AI Collaboration: A Review and Methodological Framework.
- Ge, Yuyao; Mei, Lingrui; Duan, Zenghao; Li, Tianhao; Zheng, Yujia; Wang, Yiwei et al. (2025): A Survey of Vibe Coding with Large Language Models.
- Hermann, Felix; Max, Paul Christian; Kunz, Matthias (2024): Automatisierte, BIM-basierte Ökobilanzierung am Beispiel des Infrastrukturbaus. In *Bautechnik* 101 (2), pp. 128–133. DOI: 10.1002/bate.202300115.
- Horn, Rafael; Ebertshäuser, Sebastian; Di Bari, Roberta; Jorgji, Olivia; Traunspurger, René; Both, Petra von (2020): The BIM2LCA Approach: An Industry Foundation Classes (IFC)-Based Interface to Integrate Life Cycle Assessment in Integral Planning. In *Sustainability* 12 (16), p. 6558. DOI: 10.3390/su12166558.
- Kaus, Michelle; Barkmann, Emilia (2025): Tagungsband 34. BBB Assistent*innen-Tagung 2025 Berlin. Chancen und Herausforderungen von digitalen Materialpässen für Infrastrukturprojekte. With assistance of Technische Universität Berlin, Felix Ehrlich, Sören Sommerfeld, Sophie Heinrich, Sascha Peter, Jonathan Schäfer et al.
- Meleková, A.; Kvet, M. (2026): Choosing the Right Data Format for Storing and Manipulating Database Data in HashMap. In *Lecture Notes in Networks and Systems* (1524 LNNS), pp. 153–165. Available online at https://www.scopus.com/inward/record.uri?eid=2-s2.0-105020702102&doi=10.1007%2f978-3-031-97799-2_14&partnerID=40&md5=1ebe54d2c4db9c8dc5e073b14434b590.
- Meske, Christian; Hermanns, Tobias; Weiden, Esther; Loser, Kai-Uwe; Berger, Thorsten (2025): Vibe Coding as a Reconfiguration of Intent Mediation in Software Development: Definition, Implications, and Research Agenda.
- Meyes, Richard; Lu, Melanie; Waubert de Puiseau, Constantin; Meisen, Tobias (2019): Ablation Studies in Artificial Neural Networks.
- Michael Sulzbach (n.d.): Nachhaltigkeit im Straßenbau. Hinweise zur Verwendung von Kriterien und Daten bei einer vereinfachten Ökobilanz von Straßenbaumaßnahmen, Teil 1: Gebundene Schichten/Treibhausgase/A1-A4 (H vÖkobilanz-1) – (2025) – 460/1. Edited by Forschungsgesellschaft für Straßen- und Verkehrswesen e. V. (FGSV). Available online at <https://www.fgsv.de/netzwerk/gremien/ag-4-infrastrukturmanagement/46-nachhaltigkeit-im-strassenbau>, checked on 1/16/2026.
- Oliver Fischer; Julia Sauer; Jörg Jungwirth; Ulrich Baumgärtner; Rainer Hess; Carolin Roth; Sonja Xalter (2016): Einheitliche Bewertungskriterien für Elemente der Straßenverkehrsinfrastruktur im Hinblick auf Nachhaltigkeit – Straße und Tunnel. S 97. Edited by Bundesanstalt für Straßenwesen (Berichte der Bundesanstalt für Strassenwesen).
- Petrosa, Dino; Haverkamp, Pamela; Backes, Jana Gerta; Crampen, David; Blankenbach, Jörg; Traverso, Marzia (2025): Development of a BIM-based AI-driven matching tool for LCA datasets. In *Discov Sustain* 6 (1). DOI: 10.1007/s43621-025-02203-8.
- Potrc Obrecht, Tajda; Röck, Martin; Hoxha, Endrit; Passer, Alexander (2020): BIM and LCA Integration: A Systematic Literature Review. In *Sustainability* 12 (14), p. 5534. DOI: 10.3390/su12145534.
- Rainer Hess, Martina Lohmeier, Torsten Mielecke, Katharina Kunz (2022): Nachhaltigkeitsbewertung von Streckenzügen der Straßeninfrastruktur. V361. Edited by Bundesanstalt für Straßenwesen (Berichte der Bundesanstalt für Strassenwesen).
- RWTH Aachen University (2025): SusInfra-Sustainability in Infrastructure. Rahmendokument. Edited by Mobilithek. Available online at <https://mobilithek.info/offers/755129848365465600>, checked on 1/16/2026.
- Wastiels, L.; Decuypere, R. (2019): Identification and comparison of LCA-BIM integration strategies. In *IOP Conf. Ser.: Earth Environ. Sci.* 323 (1), p. 12101. DOI: 10.1088/1755-1315/323/1/012101